November 8, 2026

Editors-in-Chief
Journal of Statistical Physics

Dear Editors,

Please find enclosed my manuscript, "Exact stationary majority fraction in group-flip consensus dynamics on quenched hypergraph balls", which I submit for consideration as an original research article in the *Journal of Statistical Physics*. The manuscript was previously submitted to *Physica A* (PHYSA-264666), where the editor declined it on grounds of scope without external review.

The manuscript studies two-colour group-flip dynamics on the quenched metric balls of a fixed hypergraph. Existing mean-field treatments of consensus on hypergraphs rely on annealed ensembles, in which geometry enters only through hyperedge sizes; here I ask which geometric quantities control the stationary state when the interaction sets are fixed. The main contributions are:

1. An exact stationary majority fraction, $E[M] = (2-q)/(3-2q)$ with $q = \beta/n$, when every ball is a strict majority and every residual is erased by a single fire, together with a rotation identity and a two-point law for the cycle.

2. Exact closed forms for two containment observables on one-dimensional hypergrids, confirmed by exhaustive enumeration.

3. Exact stationary solutions of the chain below the high-coverage threshold for all 17 one-dimensional cases with $n \leq 33$ (six as exact rationals), all below the band curve and above an exactly solved annealed analogue; the general inequality is stated as a conjecture, without proof.

4. An exploratory regression on four graph families, reported as a negative result with its limitations.

The work falls within the scope of the *Journal of Statistical Physics*, which publishes original papers in all areas of statistical physics, including collective phenomena in physical systems: it gives rigorous, exact results for the stationary state of an interacting stochastic particle system, derived through an exact conditional identity and a renewal–reward argument, together with exact finite-size solutions. The status of each result is kept explicit (theorem, conjecture, or exploratory protocol), and the manuscript states that the dynamics depends on the hypergraph only through its 2-section graph. Complete proofs are provided in the Supplementary Material (Online Resource 1).

All code and data are archived at Zenodo (`https://doi.org/10.5281/zenodo.24050030`). AI-assisted tools (Claude Opus 5.5, Claude Sonnet 5.5 and Qwen 3.8 Max) were used for code development, manuscript editing and literature search, as declared in the manuscript; the author takes full responsibility for the content.

This manuscript is original, has not been published previously, has not been posted as a preprint, and is not under consideration elsewhere. The author has no relevant financial or non-financial interests to disclose.

Thank you for your time and consideration.

Sincerely,

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